

Cambridge International AS & A Level

CANDIDATE
NAME

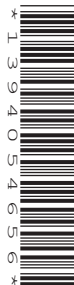
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FURTHER MATHEMATICS

9231/32

Paper 3 Further Mechanics

October/November 2023

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

INFORMATION

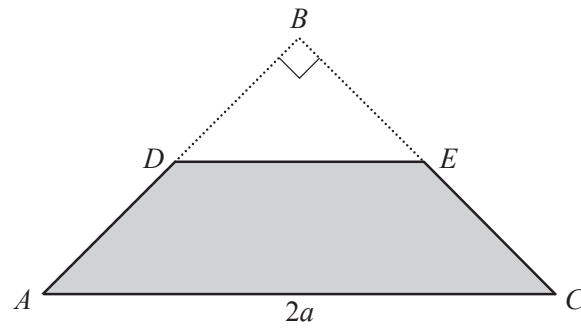
- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

[4]

[illegible]

Find v in terms of x , giving your answer in the form $v = \frac{Ax+B}{(x+1)}$, where A and B are constants to be determined. [6]



A uniform lamina is in the form of an isosceles triangle ABC in which $AC = 2a$ and angle $ABC = 90^\circ$. The point D on AB is such that the ratio $DB:AB = 1:k$. The point E on CB is such that DE is parallel to AC . The triangle DBE is removed from the lamina (see diagram).

- (a) Find, in terms of k , the distance of the centre of mass of the remaining lamina $ADEC$ from the midpoint of AC . [4]

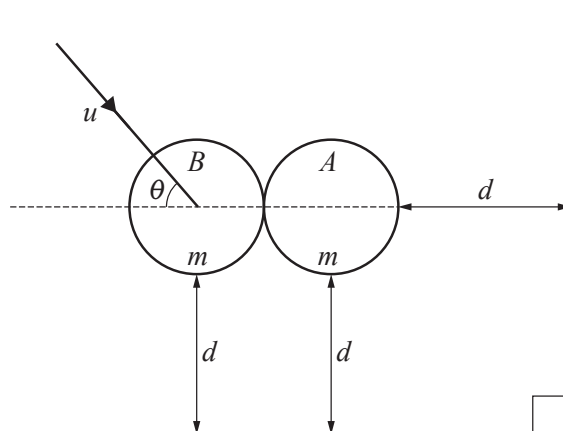
This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

When the lamina $ADEC$ is freely suspended from the vertex A , the edge AC makes an angle θ with the downward vertical, where $\tan \theta = \frac{5}{18}$.

(b) Find the value of k .

[3]

[illegible]

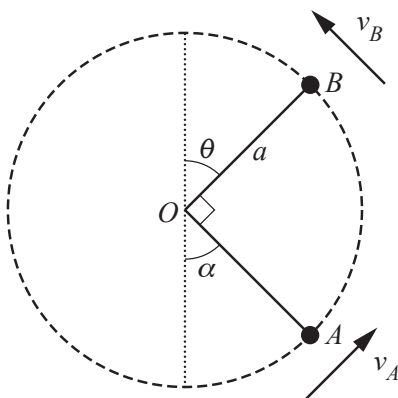


(a) Find the value of $\tan \theta$. [4]

[illegible]

- (b)** Find the percentage loss in the total kinetic energy of the spheres as a result of this collision. [3]

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for letter height. The entire page is otherwise blank, with no margins, text, or other markings.



(a) Find, in terms of m and g , the magnitude of the reaction at B . [6]

[illegible]

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

- 6 A particle P is projected with speed u at an angle α above the horizontal from a point O on a horizontal plane and moves freely under gravity. The horizontal and vertical displacements of P from O at a subsequent time t are denoted by x and y respectively.

(a) Derive the equation of the trajectory of P in the form

$$y = x \tan \alpha - \frac{gx^2}{2u^2} \sec^2 \alpha. \quad [3]$$

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During its flight, P must clear an obstacle of height h m that is at a horizontal distance of 32 m from the point of projection. When $u = 40\sqrt{2} \text{ m s}^{-1}$, P just clears the obstacle. When $u = 40 \text{ m s}^{-1}$, P only achieves 80% of the height required to clear the obstacle.

(b) Find the two possible values of h . [6]

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- (b) Find F in terms of m and g . [2]

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- (c) Find, in terms of m and g , the tension in the rod immediately before it is released. [2]

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[illegible]

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